

Week 6 — Lagrange Multipliers and KKT Conditions

Review: Equality-Constrained Maximization

Let

$$X \subseteq \mathbb{R}^J$$

be open. Let

$$f : X \rightarrow \mathbb{R}$$

be the objective function, and let

$$g_m : X \rightarrow \mathbb{R}, \quad m = 1, \dots, M,$$

be equality constraints.

We consider the problem

$$\max_{x \in X} f(x)$$

subject to

$$g_m(x) = 0, \quad m = 1, \dots, M.$$

Equivalently, define the constraint map

$$g : X \rightarrow \mathbb{R}^M, \quad g(x) = \begin{pmatrix} g_1(x) \\ \vdots \\ g_M(x) \end{pmatrix}.$$

Then the feasible set is

$$C = \{x \in X : g(x) = 0\}.$$

The Jacobian of the Constraint Map

Suppose each g_m is differentiable.

The Jacobian matrix of the constraint map g at x is

$$Dg(x) = \begin{pmatrix} \frac{\partial g_1}{\partial x_1}(x) & \cdots & \frac{\partial g_1}{\partial x_J}(x) \\ \vdots & \ddots & \vdots \\ \frac{\partial g_M}{\partial x_1}(x) & \cdots & \frac{\partial g_M}{\partial x_J}(x) \end{pmatrix}.$$

Equivalently,

$$Dg(x) = \begin{pmatrix} \nabla g_1(x)^\top \\ \vdots \\ \nabla g_M(x)^\top \end{pmatrix} \in \mathbb{R}^{M \times J}.$$

Therefore,

$$\text{rank } Dg(x) \leq \min\{M, J\}.$$

Definition (Equality constraint qualification). We say that the equality constraints satisfy the *constraint qualification* at x^* if

$$\text{rank } Dg(x^*) = M.$$

Equivalently, the constraint gradients

$$\nabla g_1(x^*), \dots, \nabla g_M(x^*)$$

are linearly independent.

Intuition. The full-rank condition says that the active equality constraints impose locally independent restrictions. No constraint is locally redundant with the others.

Geometrically, each gradient $\nabla g_m(x^*)$ is normal to its corresponding constraint surface. Full rank requires these normal directions to be linearly independent.

Lagrange Multiplier Theorem

Theorem 1 (Lagrange multiplier theorem). *Let $X \subseteq \mathbb{R}^J$ be open and suppose*

$$f, g_1, \dots, g_M : X \rightarrow \mathbb{R}$$

are continuously differentiable.

Suppose $x^ \in X$ is a local solution to*

$$\max_{x \in X} f(x)$$

subject to

$$g_m(x) = 0, \quad m = 1, \dots, M.$$

If

$$\text{rank } Dg(x^*) = M,$$

then there exist unique scalars

$$\lambda_1^*, \dots, \lambda_M^* \in \mathbb{R}$$

such that

$$\boxed{\nabla f(x^*) = \sum_{m=1}^M \lambda_m^* \nabla g_m(x^*)}.$$

Remark. The multipliers are unique here because

$$\text{rank } Dg(x^*) = M$$

implies that the vectors

$$\nabla g_1(x^*), \dots, \nabla g_M(x^*)$$

are linearly independent.

Intuition. At the constrained optimum, there is no feasible first-order direction in which the objective can increase.

Therefore the gradient of the objective must lie in the linear span of the normal vectors generated by the equality constraints:

$$\nabla f(x^*) \in \text{span}\{\nabla g_1(x^*), \dots, \nabla g_M(x^*)\}.$$

The multipliers λ_m^* are precisely the coefficients of that linear combination.

The Lagrangian with Multiple Equality Constraints

Define the Lagrangian

$$\mathcal{L} : X \times \mathbb{R}^M \rightarrow \mathbb{R}$$

by

$$\mathcal{L}(x, \lambda) = f(x) - \sum_{m=1}^M \lambda_m g_m(x).$$

Here

$$\lambda = (\lambda_1, \dots, \lambda_M).$$

The first-order conditions are

$$\frac{\partial \mathcal{L}}{\partial x_i} = \frac{\partial f}{\partial x_i}(x) - \sum_{m=1}^M \lambda_m \frac{\partial g_m}{\partial x_i}(x) = 0, \quad i = 1, \dots, J,$$

together with

$$\frac{\partial \mathcal{L}}{\partial \lambda_m} = -g_m(x) = 0, \quad m = 1, \dots, M.$$

In vector notation,

$$\nabla f(x) - \sum_{m=1}^M \lambda_m \nabla g_m(x) = 0$$

and

$$g_m(x) = 0, \quad m = 1, \dots, M.$$

Thus the “cookie-cutter” procedure is:

1. Define

$$\mathcal{L}(x, \lambda) = f(x) - \sum_{m=1}^M \lambda_m g_m(x).$$

2. Solve

$$\nabla_x \mathcal{L}(x, \lambda) = 0.$$

3. Impose

$$g_m(x) = 0, \quad m = 1, \dots, M.$$

4. Examine the resulting candidates to determine which, if any, solve the original optimization problem.

Potential Problems with the Lagrange Conditions

The Lagrange multiplier theorem gives *necessary* conditions under a constraint qualification. It does not say that every solution of the Lagrange equations is an optimizer.

There are several possible difficulties.

Problem 1: A Lagrange Critical Point Need Not Be a Maximizer

Consider

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}, \quad f(x_1, x_2) = x_1 + x_2,$$

subject to

$$g(x_1, x_2) = x_1^2 + x_2^2 - 2 = 0.$$

The feasible set is the circle

$$x_1^2 + x_2^2 = 2.$$

We have

$$\nabla f(x_1, x_2) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \nabla g(x_1, x_2) = \begin{pmatrix} 2x_1 \\ 2x_2 \end{pmatrix}.$$

The Lagrange condition is

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \lambda \begin{pmatrix} 2x_1 \\ 2x_2 \end{pmatrix}.$$

Therefore

$$1 = 2\lambda x_1, \quad 1 = 2\lambda x_2,$$

which implies

$$x_1 = x_2.$$

Using feasibility,

$$x_1^2 + x_2^2 = 2,$$

we obtain

$$2x_1^2 = 2,$$

so

$$x_1 = x_2 = \pm 1.$$

Hence the Lagrange critical points are

$$(x_1^*, x_2^*, \lambda^*) = \left(1, 1, \frac{1}{2}\right)$$

and

$$(x_1^*, x_2^*, \lambda^*) = \left(-1, -1, -\frac{1}{2}\right).$$

However,

$$f(1, 1) = 2, \quad f(-1, -1) = -2.$$

Thus

$$(1, 1)$$

is the constrained maximum, while

$$(-1, -1)$$

is the constrained minimum.

Remark. Both points satisfy the Lagrange first-order conditions. Therefore solving the Lagrange system only produces *candidates*; it does not by itself identify the maximizer.

Problem 2: The Constraint Qualification May Fail

Consider

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}, \quad f(x_1, x_2) = -x_1,$$

subject to

$$g(x_1, x_2) = x_1^3 - x_2^2 = 0.$$

Feasibility implies

$$x_1^3 = x_2^2 \geq 0,$$

so

$$x_1 \geq 0.$$

Since the objective is

$$f(x_1, x_2) = -x_1,$$

the maximum is attained at

$$x^* = (0, 0).$$

However,

$$\nabla g(x_1, x_2) = \begin{pmatrix} 3x_1^2 \\ -2x_2 \end{pmatrix},$$

and therefore

$$\nabla g(0, 0) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Thus

$$\text{rank } Dg(0, 0) = 0,$$

so the equality constraint qualification fails.

Moreover,

$$\nabla f(0, 0) = \begin{pmatrix} -1 \\ 0 \end{pmatrix},$$

and there is no scalar λ satisfying

$$\nabla f(0, 0) = \lambda \nabla g(0, 0).$$

Remark. The point $x^* = (0, 0)$ is genuinely the constrained maximizer, but the Lagrange conclusion fails because the constraint qualification fails.

Thus the regularity assumption in the Lagrange multiplier theorem is essential.

Problem 3: Lagrange Critical Points May Exist Even When No Maximizer Exists

Consider

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}$$

given by

$$f(x_1, x_2) = \frac{1}{3}x_1^3 - \frac{3}{2}x_1^2 + 2x_1,$$

subject to

$$g(x_1, x_2) = x_1 - x_2 = 0.$$

The constraint implies

$$x_2 = x_1.$$

Hence the constrained problem reduces to maximizing

$$h(x_1) = \frac{1}{3}x_1^3 - \frac{3}{2}x_1^2 + 2x_1.$$

Its derivative is

$$h'(x_1) = x_1^2 - 3x_1 + 2 = (x_1 - 1)(x_1 - 2).$$

For sufficiently large x_1 ,

$$h'(x_1) > 0,$$

and in fact

$$h(x_1) \rightarrow +\infty \quad \text{as } x_1 \rightarrow +\infty.$$

Therefore the constrained maximization problem has no solution.

Nevertheless, the Lagrange equations have solutions.

Indeed,

$$\nabla f(x_1, x_2) = \begin{pmatrix} x_1^2 - 3x_1 + 2 \\ 0 \end{pmatrix}, \quad \nabla g(x_1, x_2) = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

The condition

$$\nabla f = \lambda \nabla g$$

gives

$$x_1^2 - 3x_1 + 2 = \lambda,$$

$$0 = -\lambda.$$

Hence

$$\lambda = 0$$

and

$$x_1^2 - 3x_1 + 2 = 0,$$

so

$$x_1 \in \{1, 2\}.$$

Feasibility then gives

$$x_2 = x_1.$$

Thus the Lagrange system produces the critical candidates

$$(1, 1) \quad \text{and} \quad (2, 2),$$

even though the original maximization problem has no global maximizer.

Intuition. First-order conditions identify stationary candidates. They do not establish existence of an optimum.

If existence is known separately—for example from compactness of the feasible set and continuity of the objective—and the constraint qualification holds, then the Lagrange conditions become much more useful: we know that an optimizer exists and must appear among the Lagrange candidates.

Inequality-Constrained Maximization

We now consider

$$X \subseteq \mathbb{R}^J$$

open, with objective

$$f : X \rightarrow \mathbb{R}$$

and inequality constraints

$$g_m : X \rightarrow \mathbb{R}, \quad m = 1, \dots, M.$$

We adopt the standard-form convention

$$\boxed{g_m(x) \leq 0, \quad m = 1, \dots, M.}$$

Thus the problem is

$$\boxed{\max_{x \in X} f(x) \quad \text{subject to} \quad g_m(x) \leq 0, \quad m = 1, \dots, M.}$$

Remark (Sign convention). The choice $g_m(x) \leq 0$ is only a convention.

For example, a constraint

$$\tilde{g}_m(x) \geq b_m$$

can be rewritten as

$$g_m(x) = b_m - \tilde{g}_m(x) \leq 0.$$

Likewise, the nonnegativity constraint

$$x_i \geq 0$$

can be written as

$$g_i(x) = -x_i \leq 0.$$

Binding and Slack Constraints

Definition (Binding constraint). A constraint $g_m(x) \leq 0$ is *binding* or *active* at x if

$$g_m(x) = 0.$$

Definition (Slack constraint). A constraint $g_m(x) \leq 0$ is *slack* or *inactive* at x if

$$g_m(x) < 0.$$

Definition (Active set). The set of constraints binding at x is

$$\text{Bind}(x) = \{m \in \{1, \dots, M\} : g_m(x) = 0\}.$$

Intuition. A slack constraint does not locally restrict movement around the candidate solution. A binding constraint does: at first order, its gradient identifies a normal direction to the boundary of the feasible region.

Geometry with Two Binding Inequality Constraints

Suppose two constraints

$$g_1(x) \leq 0, \quad g_2(x) \leq 0$$

are binding at x^* .

Then

$$g_1(x^*) = g_2(x^*) = 0.$$

The gradients

$$\nabla g_1(x^*), \quad \nabla g_2(x^*)$$

are normal to their corresponding constraint boundaries and point toward directions in which the respective constraint increases—that is, toward the locally infeasible side.

At an optimum, the objective gradient must lie in the cone generated by these active constraint gradients.

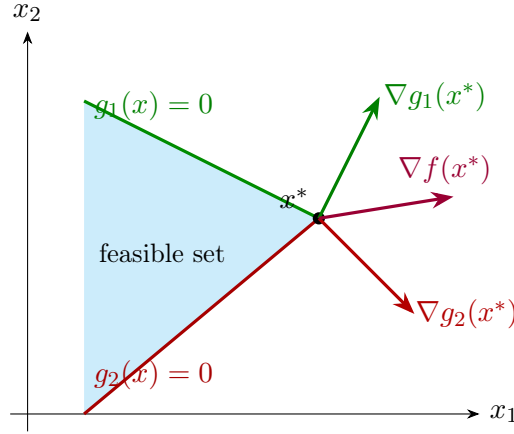


Figure 1: At a corner where two inequality constraints bind, the objective gradient must lie in the cone generated by the normals of the active constraints.

Geometrically, the relevant condition is therefore of the form

$$\nabla f(x^*) = \lambda_1^* \nabla g_1(x^*) + \lambda_2^* \nabla g_2(x^*)$$

with

$$\lambda_1^*, \lambda_2^* \geq 0.$$

More generally,

$$\nabla f(x^*) \in \text{cone} \{ \nabla g_m(x^*) : m \in \text{Bind}(x^*) \}.$$

Mangasarian–Fromovitz Constraint Qualification

To obtain a KKT theorem, we again need a regularity condition.

Definition (Mangasarian–Fromovitz constraint qualification). Let

$$g_m(x) \leq 0, \quad m = 1, \dots, M.$$

The *Mangasarian–Fromovitz constraint qualification* (MFCQ) holds at x^* if there exists a direction

$$v \in \mathbb{R}^J$$

such that

$$\boxed{\nabla g_m(x^*)^\top v < 0 \quad \text{for every } m \in \text{Bind}(x^*).}$$

Intuition. MFCQ requires the existence of a direction that moves strictly toward the interior of *all* currently binding inequality constraints.

Since the feasible side is $g_m(x) \leq 0$, the condition

$$\nabla g_m(x^*)^\top v < 0$$

means that a small movement in direction v makes the active constraint strictly negative.

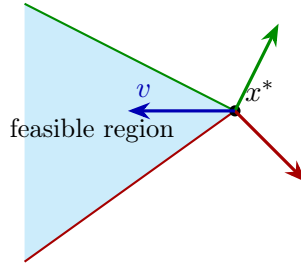


Figure 2: MFCQ asks for a direction v that moves strictly into every active inequality constraint, so that $\nabla g_m(x^*)^\top v < 0$ for every active m .

Two Simple Ways MFCQ Can Fail

Remark. Suppose g_m binds at x^* .

If

$$\nabla g_m(x^*) = 0,$$

then for every v ,

$$\nabla g_m(x^*)^\top v = 0,$$

so it is impossible to obtain the strict inequality required by MFCQ.

Hence MFCQ fails.

Remark. Suppose two constraints g_m and g_n bind at x^* and

$$\nabla g_m(x^*) = -\nabla g_n(x^*).$$

MFCQ would require

$$\nabla g_m(x^*)^\top v < 0$$

and simultaneously

$$\nabla g_n(x^*)^\top v < 0.$$

But the second condition is

$$-\nabla g_m(x^*)^\top v < 0,$$

or equivalently

$$\nabla g_m(x^*)^\top v > 0,$$

which is impossible.

Therefore MFCQ fails.

Karush–Kuhn–Tucker Theorem

Theorem 2 (KKT necessary conditions). *Let*

$$X \subseteq \mathbb{R}^J$$

be open, and suppose

$$f, g_1, \dots, g_M : X \rightarrow \mathbb{R}$$

are continuously differentiable.

Consider

$$\max_{x \in X} f(x)$$

subject to

$$g_m(x) \leq 0, \quad m = 1, \dots, M.$$

Suppose x^ is a local solution and MFCQ holds at x^* .*

Then there exist multipliers

$$\lambda_1^*, \dots, \lambda_M^* \geq 0$$

such that:

1. Stationarity

$$\nabla f(x^*) = \sum_{m=1}^M \lambda_m^* \nabla g_m(x^*).$$

2. Primal feasibility

$$g_m(x^*) \leq 0, \quad m = 1, \dots, M.$$

3. Dual feasibility

$$\lambda_m^* \geq 0, \quad m = 1, \dots, M.$$

4. Complementary slackness

$$\lambda_m^* g_m(x^*) = 0, \quad m = 1, \dots, M.$$

Complementary Slackness

Complementary slackness,

$$\lambda_m^* g_m(x^*) = 0,$$

connects the multiplier to whether a constraint actually binds.

For every m , at least one of the two terms must be zero.

If the constraint is slack,

$$g_m(x^*) < 0,$$

then necessarily

$$\boxed{\lambda_m^* = 0.}$$

Thus inactive constraints receive zero weight in the stationarity condition.

Conversely, if

$$\lambda_m^* > 0,$$

then necessarily

$$\boxed{g_m(x^*) = 0.}$$

Hence a constraint with a strictly positive multiplier must bind.

Remark. A binding constraint need not necessarily have a strictly positive multiplier. It is possible to have

$$g_m(x^*) = 0 \quad \text{and} \quad \lambda_m^* = 0.$$

Complementary slackness only rules out the combination

$$g_m(x^*) < 0, \quad \lambda_m^* > 0.$$

KKT Geometry

Stationarity can be rewritten using complementary slackness as

$$\nabla f(x^*) = \sum_{m \in \text{Bind}(x^*)} \lambda_m^* \nabla g_m(x^*), \quad \lambda_m^* \geq 0.$$

Therefore

$$\boxed{\nabla f(x^*) \in \text{cone}\{\nabla g_m(x^*) : m \in \text{Bind}(x^*)\}.}$$

This is the higher-dimensional version of the geometric picture above.

Intuition. Equality constraints generate a *linear span* of normals because their multipliers may have either sign.

Inequality constraints generate a *cone* of active normals because the KKT multipliers satisfy

$$\lambda_m^* \geq 0.$$

This difference between a span and a cone is the key geometric distinction between Lagrange multipliers for equality constraints and KKT multipliers for inequality constraints.

Nonnegativity Constraints

Suppose one of the restrictions is

$$x_i \geq 0.$$

Using our standard-form convention, write

$$g_i(x) = -x_i \leq 0.$$

Then

$$\nabla g_i(x) = -e_i,$$

where e_i is the i th standard basis vector.

If the nonnegativity constraint binds,

$$x_i^* = 0,$$

its contribution to the KKT stationarity condition is

$$\lambda_i^* \nabla g_i(x^*) = -\lambda_i^* e_i.$$

If instead

$$x_i^* > 0,$$

then the constraint is slack, so complementary slackness gives

$$\lambda_i^* = 0.$$

Thus a nonnegativity constraint affects the stationarity condition only when the corresponding variable is at its boundary.

The KKT Lagrangian

For the inequality problem

$$\max_x f(x)$$

subject to

$$g_m(x) \leq 0, \quad m = 1, \dots, M,$$

define

$$\mathcal{L}(x, \lambda) = f(x) - \sum_{m=1}^M \lambda_m g_m(x).$$

Then

$$\nabla_x \mathcal{L}(x, \lambda) = \nabla f(x) - \sum_{m=1}^M \lambda_m \nabla g_m(x),$$

while

$$\frac{\partial \mathcal{L}}{\partial \lambda_m} = -g_m(x).$$

The KKT conditions can therefore be written compactly as follows.

1. Stationarity

$$\nabla_x \mathcal{L}(x^*, \lambda^*) = 0.$$

2. Primal feasibility

$$\frac{\partial \mathcal{L}}{\partial \lambda_m}(x^*, \lambda^*) = -g_m(x^*) \geq 0.$$

3. Dual feasibility

$$\lambda_m^* \geq 0.$$

4. Complementary slackness

$$\lambda_m^* \frac{\partial \mathcal{L}}{\partial \lambda_m}(x^*, \lambda^*) = 0.$$

Since

$$\frac{\partial \mathcal{L}}{\partial \lambda_m} = -g_m(x),$$

the last condition is equivalent to

$$-\lambda_m^* g_m(x^*) = 0,$$

or simply

$$\boxed{\lambda_m^* g_m(x^*) = 0.}$$

The KKT “Cookie-Cutter” Procedure

For

$$\max_x f(x)$$

subject to

$$g_m(x) \leq 0, \quad m = 1, \dots, M,$$

the standard procedure is:

1. Define

$$\mathcal{L}(x, \lambda) = f(x) - \sum_{m=1}^M \lambda_m g_m(x).$$

2. Impose stationarity:

$$\nabla_x \mathcal{L}(x, \lambda) = 0.$$

3. Impose primal feasibility:

$$g_m(x) \leq 0.$$

4. Impose dual feasibility:

$$\lambda_m \geq 0.$$

5. Impose complementary slackness:

$$\lambda_m g_m(x) = 0.$$

6. Solve the resulting system, considering which constraints bind and which are slack.

7. Finally, verify which KKT candidates actually solve the original optimization problem.